In-Class Exercise:

Personal Loan Acceptance using PCA + KNN

Purpose Build a defensible baseline, tune k nearest neighbors, test whether PCA adds value, and interpret a two dimensional PCA visualization. Your evaluation must prevent data leakage.

Learning Outcomes

- Distinguish a majority class baseline from a predictive model baseline.
- Place normalization and PCA correctly inside validation.
- Optimize hyperparameters using the same cross validation folds.
- Compare models with accuracy, precision, recall, F1 score.
- Interpret PCA Output.

Dataset

Use bank_dataset.csv. The file contains 5,000 customer records and 14 attributes. Personal Loan is the target label.

Attribute group	Attributes	Treatment for this exercise
Identifiers	ID and ZIP Code	Exclude from modeling and explain why
Continuous or count	Age Experience Income Family CCAvg Mortgage	Inspect and normalize before distance based modeling
Coded category	Education	Keep as supplied for this exercise
Binary indicators	Securities Account CD Account Online CreditCard	Keep as predictors
Target	Personal Loan	Convert to binominal and set as label

Part I Default KNN Baseline

Create the first predictive baseline without PCA. Use k=5 and do not optimize any parameter yet.

Clean data → Cross Validation → Normalize → k NN with k 5 → Performance

1. Build the process and verify that Normalize is inside the training subprocess.
2. Apply the training fold normalization to the test fold.
3. Record accuracy, precision, recall, F1 score and the confusion matrix.
4. Explain why accuracy alone may be misleading.

Model	Accuracy	Precision 1	Recall 1	F1 1
KNN k = 5	0.9608	0.9522	0.6229	0.7525

4. Only 9.6% of customers accepted, so predicting "no" for everyone already scores 0.9040 accuracy. KNN k = 5 reaches 0.9608 but its recall on class 1 is only 0.6229, missing 181 of the 480 acceptors. Confusion matrix (out of fold): TN 4505, FP 15, FN 181, TP 299.

Part II Optimize K Without PCA

Use Optimize Parameters around the same Cross Validation process. Test integer values of k from 1 through 10. Keep all folds, preprocessing, seed, and metrics unchanged.

5. Report the best k and its cross validated metrics.
6. Show the five best parameter rows sorted by the main criterion.

Rank	k	Accuracy	Recall 1	F1 1
1	1	0.9604	0.6938	0.7706
2	3	0.9604	0.6375	0.7555
3	5	0.9608	0.6229	0.7525
4	7	0.9558	0.5688	0.7114
5	2	0.9548	0.5583	0.7033

5. Best k = 1: accuracy 0.9604, precision 0.8675, recall 0.6938, F1 0.7706 (same 5 folds and seed 42).

6. Rows above are sorted by F1 on class 1. Small k wins because acceptors are only 9.6% of the data, so a large k lets the majority outvote them.

Part III Optimize PCA and KNN

Add PCA after Normalize and before k nearest neighbors. Evaluate whether dimensionality reduction improves the model or offers an acceptable tradeoff.

Parameter	Search values
k NN k	Integers 1 through 10
PCA number of components	Integers 2 through 8

7. Run the grid search using the same 5 folds and random seed as the non PCA baseline.
8. Report the best number of components and best k.
9. Record the same metrics used for the baseline.
10. Explain why PCA must be learned separately inside every training fold.

Rank	Components	k	Accuracy	Recall 1	F1 1
1	8	1	0.9568	0.6958	0.7552
2	8	3	0.9560	0.6250	0.7309
3	8	5	0.9546	0.6000	0.7159
4	8	7	0.9514	0.5583	0.6874
5	8	4	0.9496	0.5188	0.6631

7 - 9. Grid of components 2-8 by k 1-10 (70 combinations) on the same 5 folds and seed 42. Best: components = 8, k = 1, giving accuracy 0.9568, precision 0.8270, recall 0.6958, F1 0.7552. All five best rows use the maximum 8 components.

10. PCA learns its means and eigenvectors from the data. Fitting it on the full dataset would let the test rows shape the rotation, so the test fold is no longer unseen and the score is optimistically biased.

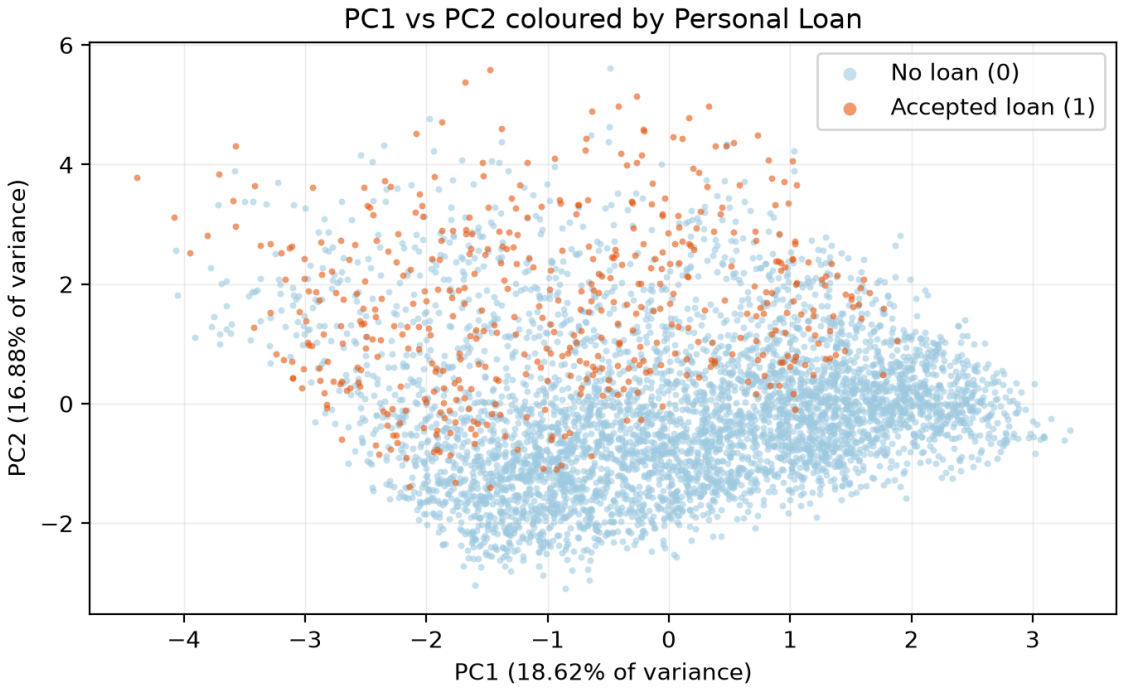
Part IV PCA Visualization and Interpretation

Create a separate descriptive workflow for visualization. Apply the agreed cleaning and attribute selection, normalize the cleaned dataset, and retain two principal components. This visualization is for interpretation; it is not an unbiased performance estimate.

11. Create a PC1 versus PC2 scatter plot and color points by Personal Loan.
12. Report the explained variance of PC1 and PC2 and their cumulative explained variance.
13. Identify the three top variable for PC1 and for PC2.

Component	Explained variance	Top variables	Interpretation
PC1	0.1862 (18.62%)	Age, Experience, Income	Career stage axis: older and more experienced, lower income
PC2	0.1688 (16.88%)	Income, CCAvg, Experience	Wealth axis: high income and high card spending

11. The PC1 versus PC2 scatter plot, coloured by Personal Loan, is shown below. 12. Explained variance: PC1 = 0.1862 (18.62%), PC2 = 0.1688 (16.88%), cumulative = 0.3550 (35.50%). 13. Top three loadings: PC1 = Age (+0.622), Experience (+0.618), Income (-0.312); PC2 = Income (+0.532), CCAvg (+0.498), Experience (+0.334).



Acceptors sit high on PC2 (mean +1.64 against -0.17) but overlap along PC1, so acceptance follows the income and spending axis, not age. Descriptive only: the scaler and PCA were fitted on all rows.

Part V Model Comparison and Decision

Model	Best settings	Dimensions	Accuracy	Recall 1
Default KNN	k = 5	Original predictors	0.9608	0.6229
Optimized KNN	k = 1	Original predictors	0.9604	0.6938
Optimized PCA KNN	components =8 and k =1	New components	0.9568	0.6958

Decision Questions

14. Which predictive model should be selected and why?
15. Did PCA improve predictive performance, preserve similar performance with fewer dimensions, or reduce performance?
16. Which metric is most important if the bank wants to identify as many potential loan customers as possible? Explain.

14. The optimized KNN with k = 1 on the original predictors: best F1 (0.7706) and better recall than the default (0.6938 against 0.6229) at the same accuracy, with no extra PCA step.

15. No. PCA with 8 components matches recall but loses accuracy, precision and F1, and the grid picked the largest component count offered, so it is a mild reduction rather than a good tradeoff.

16. Recall on class 1. Recall = TP / (TP + FN) is the share of true acceptors the model actually reaches; a missed acceptor is lost revenue while a false positive costs one wasted contact. Accuracy is unusable because 90.4% of the rows are class 0.

Part VI, apply you best model to predict a given customer

17. Predict the following.

Age	Experience	Income	Family	CCAvg	Education	Mortgage	Personal Loan	Securities Account	CD Account	Online	CreditCard
40	10	84	2	2	2	0	0	0	0	1	1

17. Personal Loan = 0, this customer is predicted not to accept. With k = 1 the nearest neighbour is a non acceptor, so P(1) = 0.00.

Required Submission

File	Required content
Process file	One XML/IPYNB containing the baseline optimized KNN and optimized PCA KNN workflows
Report	One PDF containing process screenshots settings result tables PCA plot and answers